

Bank Loan Applicant **Income Analysis Report**

Loan Applicant Income Analysis — Power BI Dashboard Report

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1. Introduction

This report is based on the "Loan Applicant Income Analysis" Power BI dashboard, which analyzes bank loan applicants' income, loan amounts, region, gender, age, and education level. Three different filtered views were captured from the dashboard, and each is analyzed in detail below.

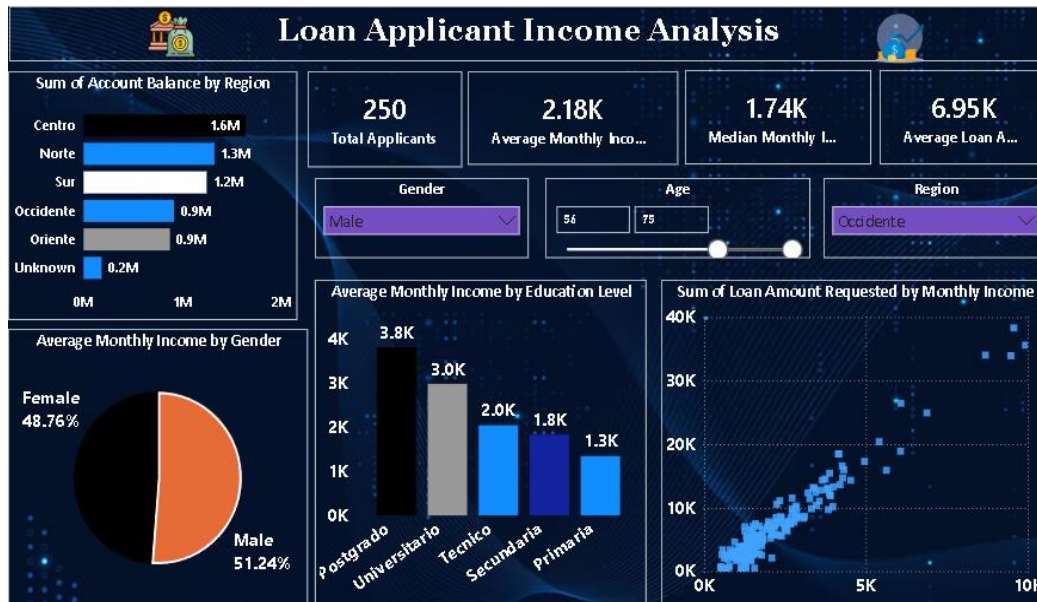
The purpose of this report is to help the bank compare income and loan patterns across different regions (Occidente, Oriente, Sur) and demographic groups, supporting better data-driven decision-making.

2. Dashboard Overview

The dashboard includes the following key visuals:

- Sum of Account Balance by Region — total account balance broken down by region
- Total Applicants, Average Monthly Income, Median Monthly Income, Average Loan Amount — key KPI cards
- Average Monthly Income by Gender — pie chart split by gender
- Average Monthly Income by Education Level — bar chart by education level
- Sum of Loan Amount Requested by Monthly Income — scatter plot of income vs. requested loan amount
- Interactive slicers for Gender, Age, and Region

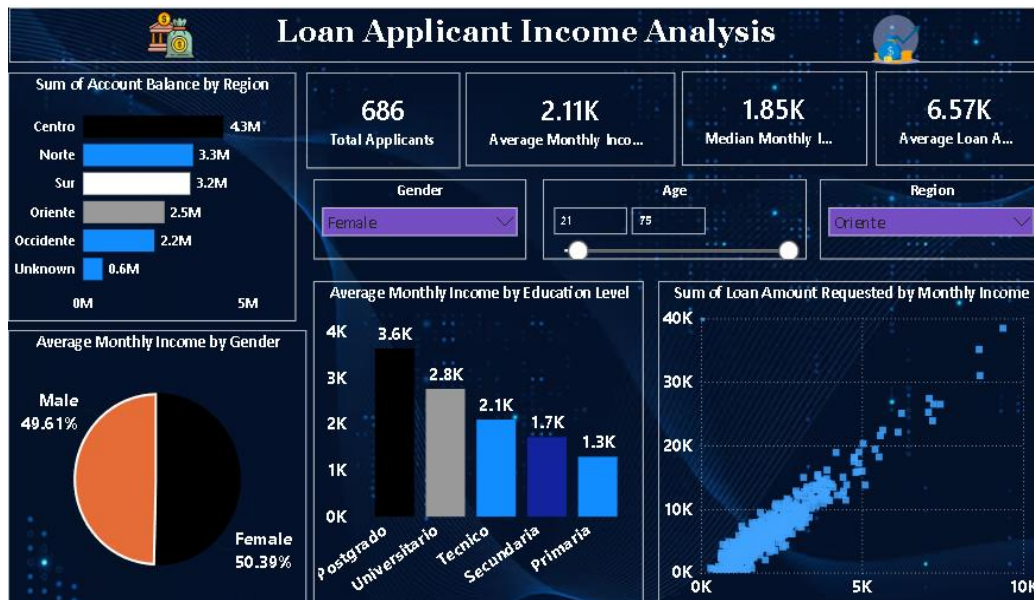
3. View 1: Male Applicants — Occidente Region, Age 56–75



This view covers 250 applicants — male, aged 56 to 75, in the Occidente region.

- Total Applicants: 250
- Average Monthly Income: 2.18K
- Median Monthly Income: 1.74K
- Average Loan Amount: 6.95K
- Account balance is highest in the Centro region (1.6M), followed by Norte (1.3M) and Sur (1.2M)
- Applicants with a Postgraduate education have the highest average income (3.8K), while Primaria (primary education) has the lowest (1.3K)

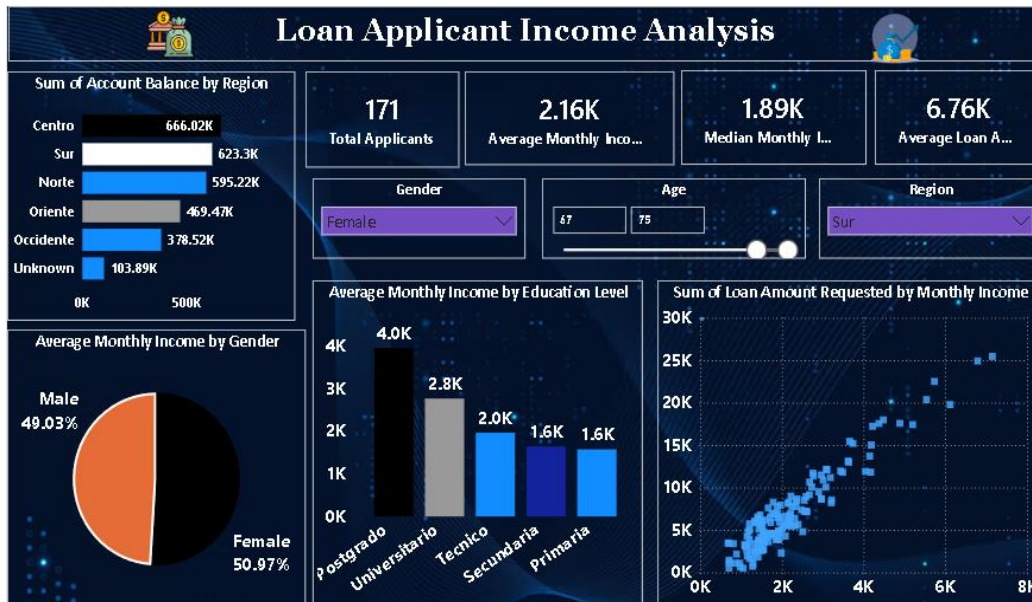
4. View 2: Female Applicants — Oriente Region, Age 21–75



This view shows female applicants in the Oriente region across all ages (21–75), with 686 applicants — the largest group among the three views.

- Total Applicants: 686
- Average Monthly Income: 2.11K
- Median Monthly Income: 1.85K
- Average Loan Amount: 6.57K
- Account balance in the Centro region is exceptionally high (43M), far larger than any other region
- The income-by-education pattern remains consistent — Postgrado highest (3.6K), Primaria lowest (1.3K)

5. View 3: Female Applicants — Sur Region, Age 67–75



This view focuses on senior (67–75 years old) female applicants in the Sur region, with the smallest group of 171 applicants — a small but distinct segment.

- Total Applicants: 171
- Average Monthly Income: 2.16K — the highest of the three views
- Median Monthly Income: 1.89K — the highest of the three views
- Average Loan Amount: 6.76K
- Postgraduate-educated applicants have the highest average income (4.0K)
- Secundaria and Primaria education levels have nearly equal income (1.6K each), a pattern different from the other views

6. Comparative Analysis

The table below compares the key metrics across all three filtered views:

Metric	View 1 (Occidente)	View 2 (Oriente)	View 3 (Sur)
Total Applicants	250	686	171
Average Income	2.18K	2.11K	2.16K
Median Income	1.74K	1.85K	1.89K
Average Loan Amount	6.95K	6.57K	6.76K
Top Region (Balance)	Centro (1.6M)	Centro (43M)	Centro (666K)
Postgrado Income	3.8K	3.6K	4.0K
Primaria Income	1.3K	1.3K	1.6K

7. Key Findings

- Across all three views, the Centro region consistently has the highest account balance — notably reaching an unusual 43M in the Oriente filter, which may indicate an outlier or a set of high-value accounts worth investigating.
- A clear positive correlation exists between education level and income — the more education (Postgrado > Universitario > Tecnico > Secundaria > Primaria), the higher the average monthly income, and this pattern holds consistently across all three views.
- The Oriente region (female, all ages) has the highest number of applicants (686), while senior applicants (67–75) in the Sur region have the fewest (171).
- Interestingly, the senior age group (Sur, 67–75) has higher average and median income than the other two groups, suggesting income tends to rise with experience/seniority.
- The Loan Amount vs. Monthly Income scatter plot shows a clear positive trend in all views — applicants with higher income tend to request larger loan amounts.

- The gender split is fairly balanced across nearly all views (roughly 49–51%), indicating near-equal male and female participation among applicants.

8. Recommendations

- Investigate the unusually high account balance in the Centro region to determine whether it reflects genuine data or is driven by a few large/outlier accounts.
- Design premium loan products targeted at highly educated customers, since their income and repayment capacity are higher.
- Create targeted, retirement-friendly loan plans for senior applicants in the Sur region, whose income is relatively stable and high.
- Since the Oriente region has the highest number of applicants, consider allocating more marketing and customer-service resources there.
- Add a time-based trend visual to the dashboard to help forecast future application patterns.

9. Conclusion

The "Loan Applicant Income Analysis" dashboard enables the bank to gain a deeper understanding of applicant income and loan patterns across region, gender, age, and education level. Analysis of the three filtered views clearly shows that education and experience (age) are key determinants of income, while regional variations — such as Centro's unusually high balance — warrant further investigation. Based on these insights, the bank can develop more targeted, data-driven lending policies.